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DATABASE SYSTEMS ENGINEERING AND DISTRIBUTED BACKEND DEVELOPMENT (25CS1302E)

Y25 - 2026-2027 Trimester - 4

Project Abstract Submission Form

Sec No-	Team No-	Project Title
7	3	SMART DELIVERY MANAGEMENT SYSTEM

1. Abstract

The Smart Delivery Management System is a web-based application designed to simplify, automate, and improve the process of managing deliveries and shipments. The system provides an efficient platform for monitoring shipment details, tracking delivery status, managing transportation routes, and maintaining delivery records.

The main objective of this project is to reduce delivery delays, improve transportation efficiency, and provide better visibility of shipment operations. It includes features such as a dashboard, shipment management, live shipment tracking, route optimization, notification management, and delivery history.

The system also focuses on efficient database management using SQL operations such as INSERT, UPDATE, DELETE, SELECT, and JOIN to manage shipment information. Indexing techniques such as B-Tree, Composite Index, and Covering Index are used to improve data retrieval and query performance.

Furthermore, the project incorporates distributed system concepts such as database replication, read scaling, sharding, and the CAP theorem to support scalability, availability, and reliable data management across multiple locations.

Overall, the Smart Delivery Management System provides a centralized and user-friendly solution for managing delivery operations. It helps logistics administrators monitor shipments, identify delays, optimize delivery routes, and maintain accurate records, making the delivery process more organized, reliable, and efficient.

Team Details

Roll Number(s) Name of the Student	Signature
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1. Frameworks / Platforms / Software Used

The following frameworks, platforms, and software tools are used in the development of the Smart Delivery Management System.

1. HTML5: Used to create the structure and layout of the web application, including dashboards, shipment details,

and tracking pages.

2. CSS3: Used to design and style the user interface with attractive layouts, colors, tables, buttons, and responsive elements.

3. JavaScript: Used to implement interactive features such as shipment tracking, navigation, searching shipment details, and displaying delivery status.

4. Python (Flask): Can be used as a backend framework to handle server-side operations, process requests, and connect the web application with the database.

5. SQL: Used to store, retrieve, update, and manage shipment information, customer details, delivery records, and tracking data.

6. Database Management System (DBMS): Used for maintaining organized shipment records and performing database operations efficiently.

7. Visual Studio Code: Used as the development environment for writing and editing HTML, CSS, JavaScript, and Python code.

8. Web Browser (Google Chrome): Used to run, test, and view the Smart Delivery Management System.

9. Distributed System Concepts: Database replication, read scaling, sharding, and the CAP theorem are used as concepts for improving scalability and reliability.

Guide Name and Signature (remarks if any)